

Practical Work 7

Design by classical filtering

Objectives:

- Familiarize with the concepts of iterative impedance, image, and loading characteristics.
- Design matched systems using concepts from classical theory.
- Design systems that allow working with impedance steps.

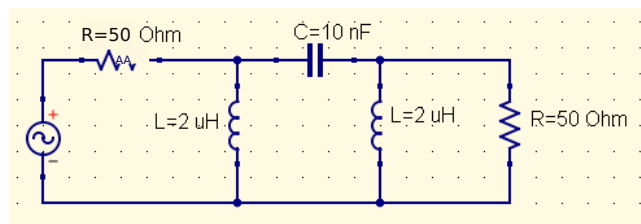
Exercise 1. Within a matched system with a characteristic impedance of 50 ohms, it is requested to design a network that achieves 20 dB attenuation. Simulate the proposed design and verify its operation.

Exercise 2. Given a PI-type high-pass network, obtain the image parameters.

Exercise 3. Design a low-pass filter with the following specifications:

- Should present matching to source and load, both $50 + j0$ ohms
- To minimize the use of inductors prototype PI cells will be used
- The prototype cell shall present infinite impedance at a frequency of 1 GHz
- The filter shall present infinite attenuation at 1.1 GHz

Exercise 4. To eliminate spurious frequencies, a constant-k filter is placed in a 50-ohm system, according to the following scheme:



It is required:

1. Determine for which frequency within the passband the filter is matched.
2. Determine the filter attenuation for that frequency
3. Same as point 1, if $R_g = R_l$ is replaced with a 1-ohm resistor
4. Determine the value of $R_g = R_l$ so that the system is matched at a frequency of 100 kHz
5. Simulate the proposed design and verify its operation.

Exercise 5. Obtain a filter with a cut-off frequency of 1 MHz that presents at 1.2 MHz an attenuation of 20 dB. Calculate the number of cells required to meet the specification assuming the cells are filters with constant K. Simulate the proposed design and verify its operation.

Exercise 6. Obtain a filter with a cut-off frequency of 1 MHz that presents at 1.2 MHz an attenuation of 20 dB. Calculate the number of cells required to meet the specification assuming the cells are m-derivative filters. Simulate the proposed design and verify its operation.

Exercise 7. Obtain a filter with a cut-off frequency at 1 kHz loaded with 600 ohms. To achieve good discrimination, it is desired to obtain an infinite attenuation frequency of some cell at 1.05 kHz. To properly adapt, sections with $m=0.6$ at the input and output are requested. Simulate the proposed design and verify its operation.

Exercise 8. Given a K-constant T-type structure, obtain the components to build a filter with the following specifications:

- Immersed in a 600-ohm system
- Cut-off frequency: 10 GHz
- Transmission zero: 9.5 GHz

Exercise 9. Design a low-pass filter with constant K with a cut-off frequency of 2 kHz loaded with its characteristic impedance of 600 ohms. Simulate the proposed design and verify its operation.

Exercise 10. Design a low-pass filter that satisfies the following specifications:

- cut-off frequency = 20K rad/s
- $R_L=600$ ohms
- Infinite attenuation at:
 - $\omega_{r1} = 22K$ rad/s
 - $\omega_{r2} = 40K$ rad/s
 - $\omega_o = \infty$
- The filter must be matched at input and output in the passband.
- Simulate the proposed design and verify its operation.

Exercise 11. Design a broadband attenuator that allows inserting 30 dB attenuation in a 75-ohm coaxial cable.

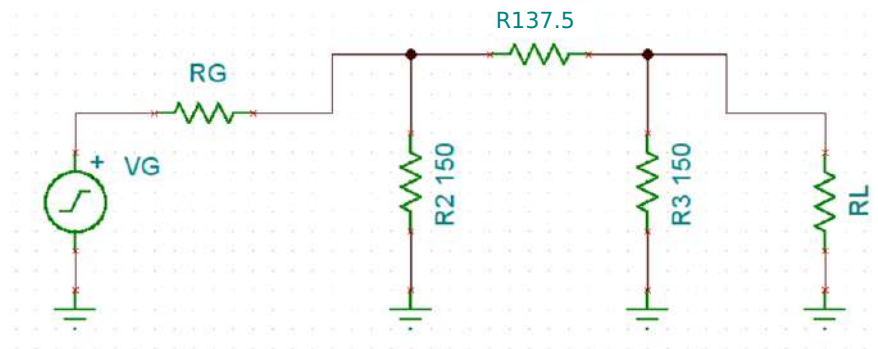
b) Verify matching, the attenuation of the resulting design.

Exercise 12. Design a broadband coupler that allows interconnection by matching two coaxial cables. Its input must allow connection of a 75 ohm characteristic impedance coaxial cable and its output one with 50 ohms characteristic impedance. It should attenuate 5.72 dB.

b) Verify the matching, the attenuation of the resulting design, and draw conclusions.

Exercise 13.

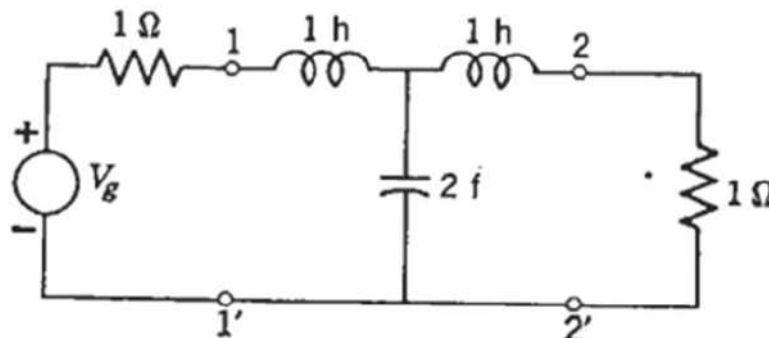
Given the following PI attenuator:



- Calculate the S parameters of the quadripole, assuming $Z_{o1}=Z_{o2}=Z_o=50$ Ohm.
- Using the S parameters, calculate the input power to the attenuator and the power dissipated in the load.
- Repeat point a) for $Z_o=75$ Ohm
- Compare the results obtained for the different values of Z_o .
- How can the S_{11}/S_{22} and S_{21}/S_{12} results obtained in both cases be interpreted?

Exercise 14.

For the following network:



- A) Calculate the S-parameters of the network.
- B) Obtain analytically and plot $|S_{11}(w)|$ and $|S_{21}(w)|$. Draw conclusions about them in stopband and attenuation band